

AI Usage Doc

All AI functions are called from backend/app/pred_engine/ai_caller.py

Should function independently from rest of system currently, except for requiring data input

API has input data formatted as it should be?*

Uses JSON files to get names, icons, etc

If None is returned no recommendation messages should be shown

Note: create_models() has to run on system startup to create all the models and store them

1. Knn Pathing Model

Used on the Match Replay pages to show the recommended path that a player should have take

Called with the function get_knn_output(data)

- data:
 - 2D array input variable
 - Values in data array :
[[x,y,teamPosition,lane,role,time,prevX,prevY,pprevX,pprevY,champExperience,champLevel,championId,gameLength,deaths,itemsPurchased,killingSprees,kills,visionScore,level,minionsKilled,timeEnemySpentControlled,xp,totalDamageDone,totalDamageDoneToChampions,totalDamageTaken,abilityHaste,abilityPower,armor,attackDamage,attackSpeed,ccReduction,cooldownReduction,health,healthMax,healthRegen,lifesteal,movementSpeed,power,powerMax,x1,y1,x2,y2,x3,y3,x4,y4,x5,y5,x6,y6,x7,y7,x8,y8,x9,y9], ...]
 - Need all the rows for the player whose path is being predicted
 - Returns: [[603.00,611.00],[5994.90,8299.70],[7014.48,7556.69],[4427.82,9752.66],[6334.94,9151.92],[4768.73,8961.70],[4289.80,10245.07],[4518.68,9368.80],[4928.24,6899.56],[4868.34,10896.21],[2129.15,3066.91],[5593.04,9868.95],[6668.33,10448.77],[6505.81,10163.79],[7026.19,8432.11],[4947.21,8168.15],[4668.29,7940.46]]

2. Champion Choice RF

Used with the Match data page to recommend a better champion for the players playstyle for the overall match

Called with the function get_champ_pred(data)

- data

- 2D array input variable
- Values in data array :
[[championId,teamPosition,role,lane,damageDealtToBuildings,damageDealtToObjective
s,damageDealtToTurrets,damageSelfMitigated,deaths,inhibitorTakedowns,inhibitorsLost
,itemsPurchased,killingSprees,kills,totalHeal,totalHealsOnTeammates,visionScore,curre
ntGold,goldPerSecond,level,minionsKilled,timeEnemySpentControlled,totalGold,xp,ma
gicDamageDone,magicDamageDoneToChampions,magicDamageTaken,physicalDamag
eDone,physicalDamageDoneToChampions,physicalDamageTaken,totalDamageDone,tot
alDamageDoneToChampions,totalDamageTaken,trueDamageDone,trueDamageDoneTo
Champions,trueDamageTaken,abilityPower,armor,armorPenPercent,attackDamage,attac
kSpeed,ccReduction,health,healthMax,healthRegen,lifesteal,magicPen,magicPenPercent
,magicResist,movementSpeed,omnivamp,power,powerMax]]
- Single row of data goes in, single champion comes out
- Returns:
 - 'Fiora'
 - None

3. Item Purchase RF

Used with the Match Replay page to recommend item purchases at different points in the game

Called with the function `get_item_pred(data)`

- data
 - 2D array input variable
 - Values in data array :
[[itemId,timestamp,lane,champExperience,champLevel,championId,currentGold,level,m
inionsKilled,timeEnemySpentControlled,totalGold,xp,x,y,magicDamageDone,magicDa
mageDoneToChampions,magicDamageTaken,physicalDamageDone,physicalDamageDo
neToChampions,physicalDamageTaken,totalDamageDone,totalDamageDoneToChampio
ns,totalDamageTaken,trueDamageDone,trueDamageDoneToChampions,trueDamageTak
en,abilityPower,armor,armorPenPercent,attackDamage,attackSpeed,ccReduction,health,
healthMax,healthRegen,lifesteal,magicPen,magicPenPercent,magicResist,movementSpe
ed,omnivamp,power,powerMax]]
 - Single row of data goes in, gives back one item name and icon url
 - Returns:
 - ('Boots', 'https://raw.communitydragon.org/16.3/plugins/rcp-be-lol-game-data/global/default/assets/items/icons2d/1001_class_t1_bootsofspeed.png')
#Name, Icon
 - None

4. Player Role RF

Used with the Match data page to recommend a better position within the team for the players playstyle for the overall match

Called with the function `get_role_pred(data)`

- data
 - 2D array input variable
 - Values in data array:
[[teamPosition, lane, championId, kills, physicalDamageDealt, totalDamagedealt, magicDamageDealt, totalHeal, totalEnemyJungleMinionsKilled, totalHealsOnTeammates, totalUnitsHealed, wardsKilled, wardsPlaced, detectorWardsPlaced, start_movementSpeed, start_health, start_healthMax, start_healthRegen, start_armor, end_movementSpeed, end_health, end_healthMax, end_healthRegen, end_armor]]
 - Takes a single row of data, gives back a position and lane value
 - Returns:
 - ('TOP', 'JUNGLE') #TeamPosition, Lane
 - None

5. Ability Upgrade RF

Used with the Match Replay page to recommend what abilities (skills) to upgrade/unlock at different points in the game

Called with the function `get_skill_pred(data)`

- data
 - 2D array input variable
 - Values in data array:
[[skillSlot, levelUpType, championId, damageSelfMitigated, deaths, kills, totalHeal, level, timeEnemySpentControlled, totalGold, xp, x, y, magicDamageDone, physicalDamageDone, totalDamageDone, totalDamageDoneToChampions, totalDamageTaken, armor, attackDamage, attackSpeed, health, healthMax, movementSpeed, power, powerMax]]
 - Takes a single row of data, returns a single skill name
 - Returns:
 - 'Bladework'
 - None